

Name: SANTIAGO HERRERO Student Number: 53052

Question:	1	2	3	4	Total
Points:	2½	2½	2½	2½	10
Score:	2.5	2.5	2.5	1	7.75

Everything must be justified.

1. Consider the following ordinary differential equation

$$\frac{dx}{dt} = 3x(t) \cos(t) \quad \text{para } t > 0, \quad (1)$$

with $x(0) = 1$.

- (a) (1 point) Use Taylor method of order 2 to solve numerically the differential equation for $t \in [0, 0.2]$. Use $\Delta t = 0.10$. Fill in the table with five digits after the decimal point.
- (b) (½ point) Repeat the previous exercise using $\Delta t = 0.05$.
- (c) (1 point) Estimate the step Δt necessary to get a final global error smaller than 10^{-6} .

t_k	$x_k^{(a)}$ $\Delta t = 0.10$	$x_k^{(b)}$ $\Delta t = 0.05$
0.00	1.00000	1.00000
0.05	XXXXXXXXXXXXXX	1.16125
0.10	1.34500	1.34803
0.15	XXXXXXXXXXXXXX	1.56374
0.20	1.80439	1.81199

2. Consider the least squares problem of adjusting data in the following table to the function
- $y = \alpha x^2 + \beta$
- .

x_k	y_k
-1	+1
+0	+2
+1	+3

- (a) (½ point) Explain clearly why the problem can be solved using QR decomposition.
- (b) (1½ points) Solve using QR decomposition.
- (c) (½ point) Find the quadratic error without computing $\|\mathbf{A}\vec{x} - \vec{b}\|_2^2$.
3. (2½ points) Consider the problem of finding the minimum of the function

$$f(x, y) = (x - y)^2 + (y - 2)^2. \quad (2)$$

Starting with $(x_0, y_0) = (0, 0)$, do two iterations of the Newton algorithm. Fill in the following table with your results. Use five digits after the decimal point.

k	x_k	y_k
0	0.00000	0.00000
1	2.00000	2.00000
2	2.00000	2.00000

4. (a) ($\frac{1}{2}$ point) Consider the matrix

$$A = \begin{pmatrix} 1 & +10 \\ +10 & -2 \\ +1 & +2 \end{pmatrix}. \quad (3)$$

How many positive singular values does A have? Justify your answer.

(b) (1 point) Calculate $\|(A)\|_2$.

(c) (1 point) Find the Lagrange-Chebyshev interpolating polynomial $p_2(x) = a_2x^2 + a_1x + a_0$ for the function $g(x) = \ln(1+x)$ in $[-0.1, +0.1]$. Fill in the following table with five digits after the decimal point.

a_2	a_1	a_0
-0.92420	1.52069	0.00000

Wrong

Question 1

a) $\frac{dx}{dt} = 3x(t) \cos(t)$, $t > 0$

$$\begin{aligned} x(t+\Delta t) &\approx x(t) + \frac{dx}{dt} \Delta t + \frac{1}{2} \frac{d^2x}{dt^2} \Delta t^2 \\ &= x(t) + 3x(t) \cos(t) \Delta t + \frac{1}{2} \left(3 \frac{dx}{dt} \cos(t) - 3x(t) \sin(t) \right) \Delta t^2 \\ &= x(t) + 3x(t) \cos(t) \Delta t + \frac{1}{2} \left(9x(t) \cos^2(t) - 3x(t) \sin^2(t) \right) \Delta t^2 \end{aligned}$$

$$X_{k+1} = X_k + 3X_k \cos(t_k) \Delta t + \frac{1}{2} X_k \left(3 \cos^2(t_k) - \sin^2(t_k) \right) \Delta t^2$$

b) Same iteration with different Δt .

Note:

- X_{k+1} is the approximation to $X(t_{k+1}) = X(0 + k \cdot \Delta t)$
- All operations were carried out directly on the calculator.

c) For the Taylor method of order 2 yields a global error which goes as $O(\Delta t^2)$.

If we estimate a bound for the global error by assuming $E \approx C \cdot \Delta t^2$

Then $E_{0.1} \approx C \cdot (0.1)^2 = \frac{C}{100}$

$$E_{0.05} \approx C \cdot (0.05)^2 = \frac{C}{400}$$

And from the table:

And assuming $E_{k+1} = X(k \cdot \Delta t) - X_{k+1} > 0$ since

both the approximations at 0.1 and 0.2 increased as Δt got smaller.

Question 2

$$a) \quad \bar{p} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \quad A = \begin{pmatrix} (-1)^2 & 1 \\ 0^2 & 1 \\ 1^2 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 1 \end{pmatrix}$$

$$\bar{y} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$$

The problem is to find $\bar{p}$ that minimizes

$$f(\bar{p}) = \|A\bar{p} - \bar{y}\|_2^2$$

The solution is $\bar{p}$ that verifies

$$A^T A \bar{p} = A^T \bar{y}$$

If A is decomposed $A = \overset{3 \times 2}{Q} \overset{3 \times 3}{R} \overset{3 \times 2}{R_1} = \overset{3 \times 2}{Q_1} \overset{2 \times 2}{R_1}$

$$(QR)^T (QR) \bar{p} = (QR)^T \bar{y}$$

$$\underbrace{R_1^T Q_1^T Q_1 R_1}_{II} \bar{p} = R_1^T Q_1^T \bar{y}$$

$$R_1^T R_1 \bar{p} = R_1^T Q_1^T \bar{y}$$

$$R_1 \bar{p} = Q_1^T \bar{y}$$

Actually, one does not begin with the normal equations.

which is valid if A is full column rank, which in this case is ~~not~~ true:

$$\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \neq k \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \quad \forall k \in \mathbb{R}$$

$$b) A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 1 \end{pmatrix} \quad \bar{q}_1 = \frac{\bar{a}_1}{\|\bar{a}_1\|_2} = \begin{pmatrix} 1/\sqrt{2} \\ 0 \\ 1/\sqrt{2} \end{pmatrix}$$

$\underbrace{\quad}_{\bar{a}_1} \quad \underbrace{\quad}_{\bar{a}_2}$

$$\bar{p}_2 = \bar{a}_2 - (\bar{a}_2^T \cdot \bar{q}_1) \bar{q}_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} - \frac{1}{\sqrt{2}} \cdot (2) \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

$$\bar{q}_2 = \frac{\bar{p}_2}{\|\bar{p}_2\|_2} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

$$\Rightarrow Q_1 = \begin{pmatrix} 1/\sqrt{2} & 0 \\ 0 & 1 \\ 1/\sqrt{2} & 0 \end{pmatrix}$$

Two are missing one column.

$$R_1 = \begin{pmatrix} \langle \bar{q}_1, \bar{a}_1 \rangle & \langle \bar{q}_1, \bar{a}_2 \rangle \\ 0 & \langle \bar{q}_2, \bar{a}_2 \rangle \end{pmatrix} = \begin{pmatrix} \sqrt{2} & \sqrt{2} \\ 0 & 1 \end{pmatrix}$$

Back to the least squares problem:

$$R_1 \bar{p} = Q_1^T \bar{y}$$

$$\begin{pmatrix} \sqrt{2} & \sqrt{2} \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \begin{pmatrix} 1/\sqrt{2} & 0 & 1/\sqrt{2} \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$$

$$\begin{pmatrix} \sqrt{2} & \sqrt{2} \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \begin{pmatrix} 2\sqrt{2} \\ 2 \end{pmatrix} \Rightarrow \boxed{\beta = 2}$$

$$\sqrt{2}\alpha + \sqrt{2} \cdot 2 = 2\sqrt{2} \Rightarrow \boxed{\alpha = 0}$$

Solution $\bar{p} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \begin{pmatrix} 0 \\ 2 \end{pmatrix}$

Question 2 - continued

$$c) E^2 = \|Q_2^T \bar{y}\|_2^2$$

$$Q = \begin{bmatrix} Q_1 & Q_2 \end{bmatrix}$$

$\begin{matrix} \text{---} 3 \times 3 \end{matrix}$
 $\begin{matrix} \text{---} 3 \times 2 & \text{---} 3 \times 1 \end{matrix}$

$$Q_2 = \bar{q}_3$$

$\bar{q}_3$ must be orthogonal to both $\bar{q}_1$ and $\bar{q}_2$.

I can compute it as the cross product of these two (and then normalize it)

$$\bar{q}_1 \times \bar{q}_2 = \det \begin{vmatrix} i & j & k \\ 1/\sqrt{2} & 0 & 1/\sqrt{2} \\ 0 & 1 & 0 \end{vmatrix} = \begin{pmatrix} -\frac{1}{\sqrt{2}} \\ 0 \\ \frac{1}{\sqrt{2}} \end{pmatrix} = \bar{q}_3$$

(is normalized)

then the error is

$$Q_2^T \bar{y} = \bar{q}_3^T \bar{y} = \left(-\frac{1}{\sqrt{2}} \ 0 \ \frac{1}{\sqrt{2}} \right) \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = \frac{2}{\sqrt{2}} = \frac{\sqrt{2}}{2}$$

$$E = \sqrt{\|Q_2^T \bar{y}\|_2^2} = \left| \frac{\sqrt{2}}{2} \right|$$

Question 3

$$f(x, y) = (x-y)^2 + (y-2)^2$$

$$\nabla f(x, y) = \left(\frac{\partial f}{\partial x} \quad \frac{\partial f}{\partial y} \right) = (2(x-y) \quad -2(x-y) + 2(y-2))$$

$$H_f(x, y) = \begin{pmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial y^2} \end{pmatrix} = \begin{pmatrix} 2 & -2 \\ -2 & 4 \end{pmatrix}$$

$$\frac{\partial^2 f}{\partial x^2} = 2 \quad \frac{\partial^2 f}{\partial y^2} = 2 + 2 = 4$$

$$\frac{\partial^2 f}{\partial x \partial y} = -2 \quad \frac{\partial^2 f}{\partial y \partial x} = -2$$

$$H_f^{-1} = \begin{pmatrix} 1 & 1/2 \\ 1/2 & 1/2 \end{pmatrix}$$

The algorithm is (Newton-Raphson):

$$\bar{x}_{k+1} = \bar{x}_k - H_f^{-1}(\bar{x}_k) \cdot \nabla f(\bar{x}_k), \quad \text{where } \bar{x}_k = \begin{pmatrix} x_k \\ y_k \end{pmatrix}$$

$$\begin{pmatrix} x_{k+1} \\ y_{k+1} \end{pmatrix} = \begin{pmatrix} x_k \\ y_k \end{pmatrix} - \begin{pmatrix} 1 & 1/2 \\ 1/2 & 1/2 \end{pmatrix} \begin{pmatrix} 2(x_k - y_k) \\ -2(x_k - y_k) + 2(y_k - 2) \end{pmatrix}$$

$$\begin{cases} x_{k+1} = x_k - [2x_k - 2y_k - x_k + y_k + y_k - 2] \\ y_{k+1} = y_k - [x_k - y_k - x_k + y_k + y_k - 2] \end{cases}$$

$$\begin{cases} x_{k+1} = +2 \\ y_{k+1} = +2 \end{cases}$$

$\Rightarrow$ There are no iterations, the algorithm converges in the first step.

Question 4

a) A has as many singular values as LI columns (the rank of the matrix) positive & non-zero

Since $\begin{pmatrix} 1 \\ 10 \\ 1 \end{pmatrix} \neq k \begin{pmatrix} 10 \\ -2 \\ 2 \end{pmatrix} \quad \forall k \in \mathbb{R}$,

then A is of rank 2 and has 2 positive non-zero singular values.

b) $\|A\|_2 = \sqrt{\max \lambda(A^T A)} = \max \sigma_i$, where σ_i are A's singular values.

~~$A \cdot A^T =$~~

$$A^T = \begin{pmatrix} 1 & 10 & 1 \\ 10 & -2 & 2 \end{pmatrix}$$

$$A^T \cdot A = \begin{pmatrix} 102 & -8 \\ -8 & 102 \end{pmatrix} \quad \text{wrong!}$$

$$\det(A^T A - \lambda I) = \det \begin{pmatrix} 102 - \lambda & -8 \\ -8 & 102 - \lambda \end{pmatrix} = (102 - \lambda)^2 - 8^2 = 0$$

$$10404 - 204\lambda + \lambda^2 - 64 = 0$$

$$\lambda^2 - 204\lambda + 10340 = 0$$

$$\lambda_1 = 94$$

$$\lambda_2 = 110$$

$$\sqrt{\max \lambda(A^T A)} = \sqrt{110} = 10.488$$

ok

$$\|A\|_2 = \sqrt{110} \approx 10.488$$

wrong!

c) $N=2$

Chebyshev nodes

$$X_l = \cos\left(\frac{2l+1}{3}\frac{\pi}{2}\right), \quad l=0,1,2$$

$$x_0 = \frac{\sqrt{3}}{2} \rightarrow y_0 = 0.62381$$

$$x_1 = 0 \rightarrow y_1 = 0$$

$$x_2 = -\frac{\sqrt{3}}{2} \rightarrow y_2 = -2.01011$$

$$L_0 = \frac{x(x + \frac{\sqrt{3}}{2})}{\frac{\sqrt{3}}{2}(\frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2})} = \frac{2}{3}x(x + \frac{\sqrt{3}}{2})$$

$$L_1 = \frac{(x - \frac{\sqrt{3}}{2})(x + \frac{\sqrt{3}}{2})}{-\frac{\sqrt{3}}{2}\frac{\sqrt{3}}{2}} = -\frac{4}{3}(x - \frac{\sqrt{3}}{2})(x + \frac{\sqrt{3}}{2})$$

$$L_2 = \frac{(x - \frac{\sqrt{3}}{2})x}{(-\frac{\sqrt{3}}{2} - \frac{\sqrt{3}}{2})(-\frac{\sqrt{3}}{2})} = \frac{2}{3}x(x - \frac{\sqrt{3}}{2})$$

$$P_2(x) = \sum_{k=0}^2 y_k L_k(x) = 0.62381 \cdot \frac{2}{3}x(x + \frac{\sqrt{3}}{2}) + 0$$

$$- 2.01011 \cdot \frac{2}{3}x(x - \frac{\sqrt{3}}{2})$$

$$P_2(x) = -0.92420 x^2 + 1.52069 x + 0$$